

Reviewer Pack — Minimal Rank of Configuration Spaces over XOR-AND Tree Width...

Entry 2a32502f51d1 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Configuration Spaces over XOR-AND Tree Width with Respect to Geometric Quantization

Entry ID: 2a32502f51d1 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-27 05:53:33 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology (Configuration Spaces) **Field B** (complexity object): Complexity Theory: XOR-AND Tree Width

Statement:

[‘For every n -ary Boolean function f , the minimal rank of the configuration space $C(S_f)$ associated with the set $S_f = \{x \in \{0,1\}^n \mid f(x) = 1\}$ is upper-bounded by the XOR-AND tree width $w(f)$.’, ‘More formally, there exists a constant $c > 0$ such that for all n -ary Boolean functions f , $\text{rank}(C(S_f)) \leq cw(f)$.’, ‘If this bound holds for all instances with $\text{rank}(C(S_f)) = \Theta(w(f))$, then it provides a direct geometric approach to bounding the complexity of XOR-AND trees.’]

Rationale (proposer’s reasoning):

[‘Configuration spaces in algebraic topology have been used to study geometric structures associated with combinatorial problems, but their application to complexity theory is rare.’, ‘The conjecture connects configuration spaces with XOR-AND tree width, a measure of the computational difficulty of evaluating Boolean functions.’, ‘If true, this would provide a novel perspective on understanding the structure of XOR-AND trees and could potentially lead to new algorithms for circuit complexity.’]

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 81321f666c73a2d2

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For each n -ary Boolean function f , if the rank of the configuration space $C(S_f)$ exceeds $cw(f)$, where c is a constant and $w(f)$ is the XOR-AND tree width, the conjecture is falsified.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	1.00	SAFE	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "configuration space" AND "XOR-AND tree width" AND algebraic topology"
- "minimal rank" AND "geometric quantization" AND boolean function" - "Boolean function"
AND "XOR-AND tree width" AND configuration space rank

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def xor_and_tree_width(f):
        n = len(f)
        if n == 1:
            return 0
        if all(f[i] == f[0] for i in range(1, n)):
            return 1
        mid = n // 2
        left_width = xor_and_tree_width(f[:mid])
        right_width = xor_and_tree_width(f[mid:])
        return max(left_width, right_width) + 1

    def configuration_space_rank(f):
        n = len(f)
        if n == 1:
            return 1
        if all(f[i] == f[0] for i in range(1, n)):
            return 1
        mid = n // 2
        left_rank = configuration_space_rank(f[:mid])
        right_rank = configuration_space_rank(f[mid:])
        return max(left_rank, right_rank) + 1

    def generate_random_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    n_values = [5, 10, 15, 20, 30, 40]
    results = []
```

```

for n in n_values:
    f = generate_random_boolean_function(n)
    width = xor_and_tree_width(f)
    rank = configuration_space_rank(f)
    results.append((n, width, rank))

total_instances = sum(1 for _, _, _ in results)
max_rank = max(rank for _, _, rank in results)
min_width = min(width for _, width, _ in results)

if max_rank > 2 * min_width:
    return {
        "metric_name": "rank_over_width",
        "metric_value": max_rank / min_width,
        "instances_tested": total_instances,
        "conjecture_holds": False,
        "counterexample": f"max_rank={max_rank}, min_width={min_width}"
    }
else:
    return {
        "metric_name": "rank_over_width",
        "metric_value": max_rank / min_width,
        "instances_tested": total_instances,
        "conjecture_holds": True,
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_dev = math.sqrt(sum((result["metric_value"] - mean_value)**2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_dev} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_dev} support_fraction={support_fraction}")
    else:
        for result in results:
            if not result["conjecture_holds"]:
                print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_fail={seed}")
                break

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means we cannot confirm or refute the conjecture with the available evidence. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11917
2	preregistration	ollama_remote	glm4:latest	0	0	5421
3	novelty	ollama_remote	glm4:latest	0	0	4718
4	novelty	ollama_remote	glm4:latest	0	0	6252
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	21301
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9306
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8679
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9368
9	judge	ollama_remote	glm4:latest	0	0	13002

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 89964 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/2a32502f51d1.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/2a32502f51d1.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/2a32502f51d1.tar.gz` (if generated)